Enhanced oxygen activation achieved by robust single chromium atom-derived catalysts in aerobic oxidative desulfurization

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ABSTRACT: Oxidative desulfurization (ODS) plays critical roles in the production of high-quality sulfur impurity-free fuels, especially procedures using molecular oxygen as the sole oxidant. However, state-of-the-art systems still rely on the noble metal nanocatalysts with deactivation issues caused by coking and sintering still present. Herein, leveraging the merits of single atom catalysts (SACs) with earth-abundant metal cores and robust nanoporous supports, a series of catalysts composed of homogeneously distributed single chromium atoms anchored on multi-walled carbon nanotubes were fabricated and deployed to catalyze the aerobic ODS transformation. Adopting aromatic sulfur compounds (ASCs)-containing alkanes as a model system with molecular oxygen as the oxidant, efficient oxygen-to-active $O_2^{\bullet-}$ transformation and subsequent ASCs-to-sulfone conversion have been achieved, with the former benefiting from the chromium active sites and the later arising from the robust, nanoporous, and π-conjugated architecture of the supports. The attractive catalytic performance and cycling stability of the chromium-based SACs make them promising candidates in practical sulfur species removal from liquid fuels, supplying an alternative guidance on the catalyst design towards cost-effective and energy-efficient ODS procedures.

KEYWORDS: *single-atom catalyst, aerobic oxidation, deep desulfurization, carbon nanotube, chromium*

Organic transformation procedures involving catalytic oxidation pathways represent one of the most widespread and critical methodologies in the fields of organic synthesis, environmental remediation, separation and so on.[1-2] Introduction of robust and active catalytic sites towards efficient activation of the oxidants is recognized as the key to success in the whole oxidative transformation procedure. Compared with the generally deployed organic/inorganic peroxides (aqueous hydrogen peroxide, *tert*-butyl hydroperoxide, and sodium hypochlorite), utilization of oxygen as the oxidant exhibits the merits in terms of preferred economic cost, less potential environmental issue, and no hazardous byproduct generation.[3-5] However, molecular oxygen is unreactive in the triplet ground state and nanocatalysts with enhanced oxygen activation performance are preferred especially under mild conditions. The state-of-the-art catalytic systems in oxygen-involved transformations are mainly focused on noble metal species (e.g., Au, Pt, and Pd) in the form of homogeneous metal complex, supported metal nanoparticles (NPs), or metal oxides.[6] Particularly, taking advantage of the special *d* orbitals and a high proportion of valence electrons in noble metal NPs-derived nanocatalysts, high catalytic efficiencies are achieved in diverse oxidative procedures.[7-9] However, the high cost and coking/sintering-induced deactivation of noble metal nanocatalysts pose a barrier to practical applications.[10] Cost-effective procedures capable of achieving noble metal-free transformation, high catalytic activity, and good long-term stability in oxygen activation and the whole oxidative transformation are highly desired.

Oxidative desulfurization (ODS) is essential to the environmental remediation of the global catastrophe caused by SO_x emission from the combustion of fuels containing sulfur impurities.[11-12] Particularly, compared with thiols and disulfides, removal of refractory aromatic sulfur compounds (ASCs) with inert chemical stability and high solubility in fuels is more challenging. In the traditional hydrodesulfurization (HDS) procedure, harsh operation conditions are required to achieve satisfied removal efficiency of ASCs (e.g., high temperature and hydrogen pressure).[13-15] Adsorptive desulfurization (ADS)[16] and extractive desulfurization (EDS)[17] pathways towards ASCs removal are encountered the issues of low efficiency and tedious operation. Comparatively, cost-effective and energy-efficient ASCs removal via ODS could be achieved by adapting nanocatalysts with good performance under mild conditions.[11-12, 18] Besides the oxidative catalytic activity of the metal cores, nanocatalyst skeletons capable of adsorbing the as-generated products are preferred to afford high-quality fuels without further purification.[19-20] Polyoxometalates (POMs) and supported noble metal species/NPs are the most widely deployed catalysts in ODS, but their applications are still limited by the complicated synthesis procedure, inferior intrinsic catalytic activity/stability, and tedious catalyst separation.[21-22] As a new frontier in heterogeneous catalysis, single atom catalysts (SACs) are known to possess unparalleled merits in precious metal thrifting, atom economy, and active site homogeneity, which exhibit improved catalytic activity and task-specific selectivity achieved in thermo-/photo-/electrocatalysis, compared with the counterparts possessing large particle sizes.[2, 9, 23-27] For example, carbon nitride supported cobalt single atoms (SAs) performed well in the ethylbenzene oxidation under air atmosphere, while the supported cobalt NPs are inert.[28] In the field of ODS, there are only few studies focusing on the SACs-promoted procedures. Defective UiO-66(Zr) anchored by single tungsten sites could promote the oxidative removal of sulfur compounds in the presence of H_2O_2, but was inactive if molecular oxygen was used as the oxidant.[29] TiO_2@nitrogen-doped graphene (NG) core-shell material supported Cu-Ni bimetallic SAs were adopted in the removal of dibenzothiophene under visible light irradiation.[30] Although intensive studies has been made in ODS due to its critical role in producing sulfur-free clean fuels,[31-32] there is still a lack of nanocatalysts that could leverage the merits of SACs especially with earth-abundant metal cores and robust nanoporous supports and achieve attractive aerobic ODS transformation in terms of efficient catalytic activity, robust stability, and attractive oxidized product adsorption capability.

Herein, a series SACs composed of homogeneously distributed single chromium atoms anchored on multi-walled carbon nanotubes were constructed via a facile wet impregnation/calcination procedure (Figure 1) (denoted as Cr/MWCNT). The catalytic performance of the as-afforded SACs in aerobic ODS transformation was evaluated adopting ASC-containing alkanes as a model system. With molecular oxygen as the sole oxidant, high removing efficiency of refractory sulfur compounds such as dibenzothiophene (DBT), 4-methyldibenzothiophene (4-DMDBT) and 4,6-dimethyldibenzothiophene (4,6-DMDBT) was achieved taking advantage of the combined merits of robust nature of the SAs derived from the metal-support interactions, high oxygen activation ability of chromium atoms to generate active $O_2^{\bullet-}$ species, and high chemical/thermal stability, ordered nanoporous channels, and π-conjugated skeletons of the support architecture. The good cycling stability and high sulfur removal efficiency make the Cr/MWCNT catalysts promising candidates in practical ODS transformation. This study supplies a facile strategy to generate sulfur-free fuels via oxygen-involved transformation deploying earth-abundant metal-derived nanocatalyst.

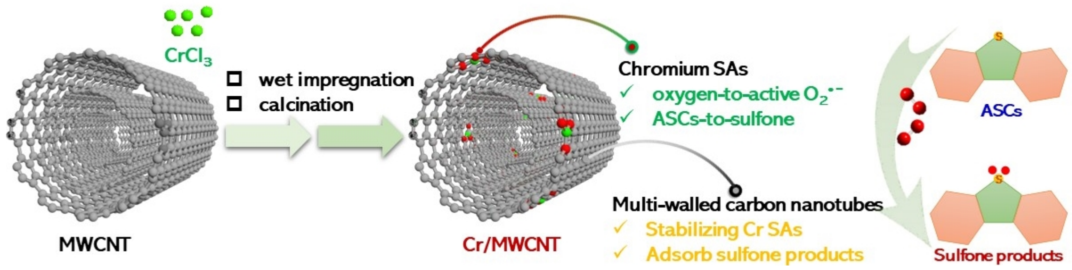

Figure 1. Schematic illustration of the Cr/MWCNT synthesis and utilization as catalyst in ODS.

SACs with low cost earth abundant metal cores have attracted more and more attention in heterogeneous catalysis.[26] Besides the active metal centers, the electronic and structural properties of the supports are critical to stabilize the metal sites during the reaction procedure, tame the local environment of the single metal atoms, and influence the adsorption/desorption behavior of the substrates/intermediates/products.[25] Towards oxidative reactions, less explored single chromium atoms supported on metal oxides exhibited good performance in methane oxidation using H_2O_2 as oxidant.[27] This envisaged us to further explore the catalytic behavior of single chromium atom-derived catalysts in more challenged oxygen activation and subsequent ODS procedure. As stated above, supports with nanoporous structures and high adsorption capacity towards the oxidative products are preferred in ODS to generate clean fuels without further extraction.[33] To test the feasibility, multi-walled carbon nanotube (MWCNT) with robust and nanoporous architecture, which has been widely used in the field of separation,[34] and extra oxygen-containing moieties on the walls for anchoring and stabilizing metal sites was deployed as the support to fabricate single chromium atoms-based catalyst.[35] The SACs were prepared by a wet impregnation and low-temperature calcination method (Figure 1). Specifically, $CrCl_3 \cdot 6H_2O$ was used as the chromium source and first supported on MWCNT (length = 10–30 μm, diameter = 20–30 nm) by wet impregnation. Then the resulted sample was annealed and calcined under nitrogen atmosphere at 400 °C for 2 h. For comparison, a series of catalysts were prepared by adding different amounts of Cr, 0.4, 0.8, 1.6, and 8 wt%, respectively, which were denoted as Cr/MWCNT-x (x = Cr amount being added). The Cr content in the as-produced catalysts was determined to be 0.13 (Cr/MWCNT-0.4), 0.36 (Cr/MWCNT-0.8), and 0.85 wt% (Cr/MWCNT-0.16), respectively, by inductively coupled plasma atomic emission spectroscopy (ICP-AES) (Table S1). Cr/MWCNT-1.6 was taken as an example to identify the structure variation of the support during the catalyst fabrication procedure and existing form of the Cr site. The porosity of Cr/MWCNT-1.6 was evaluated by N_2 adsorption/desorption curves collected at 77 K (Figure 2A). The surface area was calculated to be 178 $m^2\ g^{-1}$ using the Brunauer–Emmett–Teller (BET) method, which was a little bit higher than the original MWCNT support precursor (158 $m^2\ g^{-1}$, Figure S1), probably owing to the removal of some surface functionalities during the calcination treatment. The porous structure of MWCNT was maintained well after Cr loading, dominated by mesopores around 2.6 nm (Figure 2B and S2), indicating no big particle formation to block the porous channels within the support architecture. The crystalline structure of Cr/MWCNT-1.6 was verified by the powder X-ray diffraction (PXRD) pattern (Figure 2C and S3). Characteristic peaks being assigned to the highly crystalline graphitic nanotubes were displayed at 2θ = 26.1°, 43.1° and 53.5°, corresponding to crystal plan of (002), (100), and (004), respectively. In addition, there are no diffraction peaks for Cr_2O_3 or CrO_2 being observed in Cr/MWCNT-1.6, implying a well-dispersed feature of the Cr species. Raman spectra of the pristine MWCNT and the as-synthesized Cr/MWCNT-1.6 are shown in Figure S4. The I_D/I_G ratio of Cr/MWCNT-1.6 is lower than that of MWCNT, indicating less defects were included after metal sites introduction, which may lead to enhanced conductivity and improved electron transfer efficiency during the oxidation procedure.[36]

The morphology of Cr/MWCNT-1.6 was characterized by transmission electron microscopy (TEM) (Figure 2D), exhibiting similar nanotube structure with the MWCNT support (Figure S5). To further determine the dispersion status of Cr species on MWCNT, spherical aberration corrected high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) analysis of Cr/MWCNT-1.6 was performed (Figure 2E). Isolated Cr atoms can be clearly observed as bright dots, confirming the successful anchoring of chromium SAs on the MWCNT. However, further increasing the loading amount by adding 8 wt% Cr during the preparation of Cr/MWCNT-8 led to distinct aggregation of the Cr species (Figure 2F), which may influence the catalytic efficiency of the as-afforded materials in ODS as shown in the catalysis study. Notably, the porosity features of Cr-MWCNT-8 have no significant difference with the original MWCNT support (Figure S1, S2, and S6). X-ray absorption near-edge structure (XANES) and Fourier transform extended X-ray absorption fine structure (EXAFS) spectroscopy were deployed to further probe the chemical state and coordination structure of Cr species in Cr/MWCNT-1.6. As shown in Figure 2G, the

Cr K-edge XANES spectrum of the Cr/MWCNT-1.6 was quite different from that of Cr foil, Cr_2O_3 and CrO_2, revealing the valence state of Cr is between 0 and +3, which was a little lower than the surface Cr sites derived from the Cr2p X-ray photoelectron spectroscopy (XPS) spectrum (Figure 2I). This is probably attributed to the fact that the surface Cr atoms are more likely to be oxidized by oxygen in air.[37] Comparison of the O1s XPS spectra of MWCNT and Cr-MWCNT-1.6 revealed that the C–O bonds within the skeleton of MWCNT were leveraged to coordinate with the Cr sites and form Cr-O bonds, while the C=O bonds were intact (Figure S7). The EXAFS spectra in Figure 2H presented the dominant peaks at about 1.52 and 2.51 Å, which can be assigned to Cr–O coordination due to the absence of nitrogen element in the raw materials and no Cr–Cl bond is observed. The XPS confirmed that no peaks of Cl 2p and N 1s are observed in Cr/MWCNT-1.6 (Figure S8). Moreover, no visible Cr–Cr coordination peak was detected, which indicated no Cr clusters or nanoparticles were formed in Cr/MWCNT-1.6. The wavelet transform (WT) analysis is a powerful tool for separating backscattering atoms that provides not only a radial distance resolution, but also resolution in the k-space.[8] The intensity maxima at about 4.1 $Å^{-1}$ of Cr/MWCNT-1.6 (Figure 2J) corresponded to the Cr–O bonding. In contrast, Cr-foil, Cr_2O_3 and CrO_2 show the WT maximum at ca. 7.4, 4.5 and 4.5, respectively (Figure 2K-2M). This matched well with the EXAFS fitting results in R space. The quantitative structural parameters of Cr sites in different samples are obtained from EXAFS curve fitting (Figure S9). The corresponding EXAFS fitting parameters showed that the coordination number of center Cr atoms is about 6.1 and the average bond length of Cr SACs is 1.99

Å (Table S2). These results confirmed the existence of single Cr atoms on MWCNT, which was consistent with the HAAFT-STEM image.

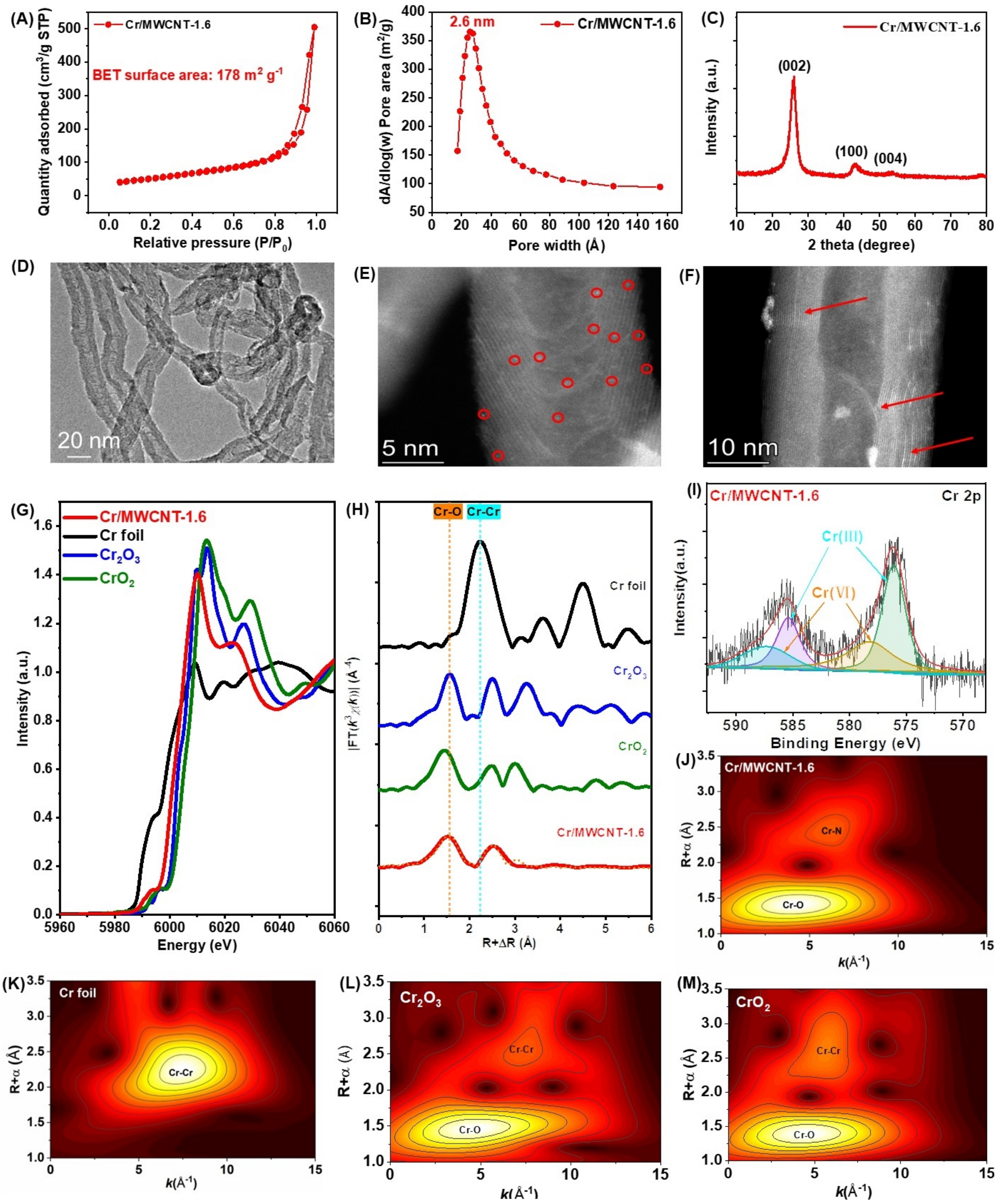

Figure 2. (A) N_2 adsorption/desorption isotherm (77 K) of Cr/MWCNT-1.6. (B) Corresponding pore size distribution curve of Cr/MWCNT-1.6. (C) PXRD pattern of Cr/MWCNT-1.6. (D) TEM and (E) HAADF-STEM image of Cr/MWCNT-1.6. (F) HAADF-STEM image of Cr/MWCNT-8. (G)The Cr K-edge XANES spectra and (H) Fourier-transform K^3-weighted EXAFS of Cr foil, Cr_2O_3, CrO_2, and Cr/MWCNT-1.6.

The orange dotted line represents the fitting of EXAFS spectra. (I) Cr 2p core-level spectrum of Cr/MWCNT-1.6. (J-M) The wavelet transforms from experimental data for Cr/MWCNT-1.6 (J), Cr foil (K), Cr_2O_3 (L), and CrO_2 (M).

The catalytic activities of as-prepared samples for ODS were assessed at 120 °C with DBT in *n*-dodecane as the model oil and molecular oxygen as an oxidant (Figure 3A). As shown in Figure 3B, either MWCNT or N_2 treated MWCNT had a really poor ODS activity (<10%), which indicated that there were no active sites produced by treating the bare MWCNT under N_2 atmosphere. Notably, the sulfur removal rate over Cr/MWCNT-1.6 with Cr content of 0.85 wt% reached 97.3% in 3 h and 100% in 5 h. Even if the Cr loading amount decreased to 0.13 wt% (Cr/MWCNT-0.4) and 0.36 wt% (Cr/MWCNT-0.8), the sulfur removal could still achieve 93.0% and > 99% within 8 h, respectively, indicating the high desulfurization efficiency being achieved by the Cr SAs-based catalysts. However, the catalyst Cr/MWCNT-8 with more Cr precursor being added was almost inactivated, probably due to the aggregation of Cr atoms, which can be observed by STEM image (Figure 2F), further demonstrating the importance of Cr SAs in oxygen activation. The porosity features of Cr/MWCNT-8 were similar to the original MWCNT support (Figure S1, S2, and S6), indicating that the diminished catalytic activity was not induced by the blocking of pores. To further reveal the catalytic activity of the Cr/MWCNT SACs, several representative catalysts using oxygen as an oxidant for ODS are summarized in Table S3. It can be observed that the turnover frequency (TOF) values of Cr/MWCNT-0.4, Cr/MWCNT-0.8 and Cr/MWCNT-1.6 reached 58.2, 25.1, and 24.8 h^{-1}, respectively, which were even higher that the noble-metal based catalysts, such as Pt/h-BN (15.3 h^{-1}) [38] and PtCu/BNNS (2.9

h^{-1}).[39] Notably, the TOF value being obtained by Cr/MWCNT-0.4 was 448 times higher than that of the Cr-based MIL-101 catalyst.[40] Therefore, the results revealed the extremely high catalytic activity of Cr SA-derived catalysts.

Subsequently, the role of the support was explored. Through the facile Cr SA-doping strategy, different carbon materials were deployed as the carriers to support Cr species with the same amount of Cr precursor being added (1.6 wt%), and the as-prepared catalysts were used to catalyze the DBT oxidation. As seen in Figure S10, within a reaction time of 3 h, 97.3% sulfur removal was achieved by Cr/MWCNT-1.6. Comparatively, when graphite was used as the support, the as-afforded Cr/graphite (Cr/Gra) exhibited inferior sulfur removal (~60%). In addition, the as prepared Cr/activated carbon (Cr/AC) delivered pretty low oxidation efficiency with sulfur removal of <10%, even inferior compared with the nonporous graphite-involved counterpart. In order to gain a deep understanding in the influence of the supports on the ODS catalytic activity being delivered by the supported Cr catalysts, detailed characterizations and comparisons were performed. HAADF-STEM images of Cr/Gra-1.6 and Cr/AC-1.6 exhibited obvious aggregation of the Cr NPs (Figure S11), which was different from the highly dispersed Cr SAs in Cr/MWCNT-1.6 (Figure 2E), indicating that the structure properties of MWCNT could benefit the coordination and stabilization of Cr SAs. Both O1s XPS and EXAFS results of the Cr/MWCNT-1.6 demonstrated that oxygen-containing species, particularly the C-O bond, in the support played critical role in SA formation by Cr-O bond construction (Figure 2 and Figure S7). The surface oxygen content of graphite obtained from the XPS survey spectra was

11.0 at.%, much higher than that in MWCNT (2.1 at.%), and O1s XPS spectra of MWCNT and graphite revealed that they possessed similar compositions of C-O and C=O functionalities (Figure S7A, S12, and S13). Boehm titration was further performed to quantify the oxygen-containing species in the bulk materials (Table S4). The -OH groups were the major contributions in both MWCNT (88%) and graphite (84%). However, porosity analysis of Cr/Gra-1.6 demonstrated its dense phase with no pores existed in the architecture. XRD pattern of Cr/Gra-1.6 exhibited good crystallinity (Figure S15), with no defects being observed in the Raman spectrum (Figure S16). Therefore, compared with Cr/MWCNT-1.6, the diminished catalytic activity of Cr/Gra-1.6 maybe caused by the inferior nonporous structures, limiting the dispersity of the Cr species in the metal loading procedure and the mass transfer during the ODS reaction. Both XPS analysis (Figure S17) and Boehm titration of AC demonstrated that the oxygen species were dominated by O-C=O and COOH, which was quite different from the case in MWCNT. The O1s XPS spectra of MWCNT and Cr/MWCNT-1.6 revealed that the C-O bond has been leveraged to coordinate with the Cr SAs (Figure S7). Therefore, the low concentration of hydroxyl groups in AC may lead to the formation of bulk Cr NPs via the wet impregnation/calcination procedure. Detailed porosity analysis displayed high surface area of Cr/AC-1.6 (1332 m^2 g^{-1}), mainly contributed by micropores (micropore surface are: 1068 m^2 g^{-1}) (Figure S18A). Pore size distribution curves of Cr/AC-1.6 further presented the micropores around 1.4 nm (Figure S18B). The abundant micropores within the architecture of AC may lead to strong adsorption of the solvent/reactant molecules, causing mass transfer limitation (particularly the gaseous O_2 in air) during the ODS procedure. In addition, compared with the highly crystalline and conjugated skeleton of MWCNT and graphite, the amorphous structure of AC, as indicated by the XRD pattern (Figure S19), was inferior in electron transfer. Therefore, compared with Cr/MWCNT-1.6, the diminished catalytic activity of Cr/AC-1.6 was possibly caused by the high content of the less preferred C=O species in AC, leading to relatively weak coordination and aggregation of the Cr NPs.

Other metal sites, including Mn, Fe, Co, and Ni, were introduced into the MWCNT skeleton via the same procedure as Cr and with the same loading amount. However, the as-afforded catalysts all exhibited diminished catalytic activity compared with that of Cr/MWCNT (Figure S20), emphasizing the excellent O_2 activation capability of Cr SAs. Therefore, the outstanding oxidation efficiency of Cr/MWCNT-1.6 in ODS was ascribed to three-fold aspects, including the abundant -OH species on the surface to facilitate the coordination and stabilization of Cr SAs, ordered porous channels in MWCNT to facilitate mass transfer, the conjugated skeletons in MWCNT to facilitate metal-support electron transfer, and the excellent O_2 activation capability of the Cr sites in the form of SAs.

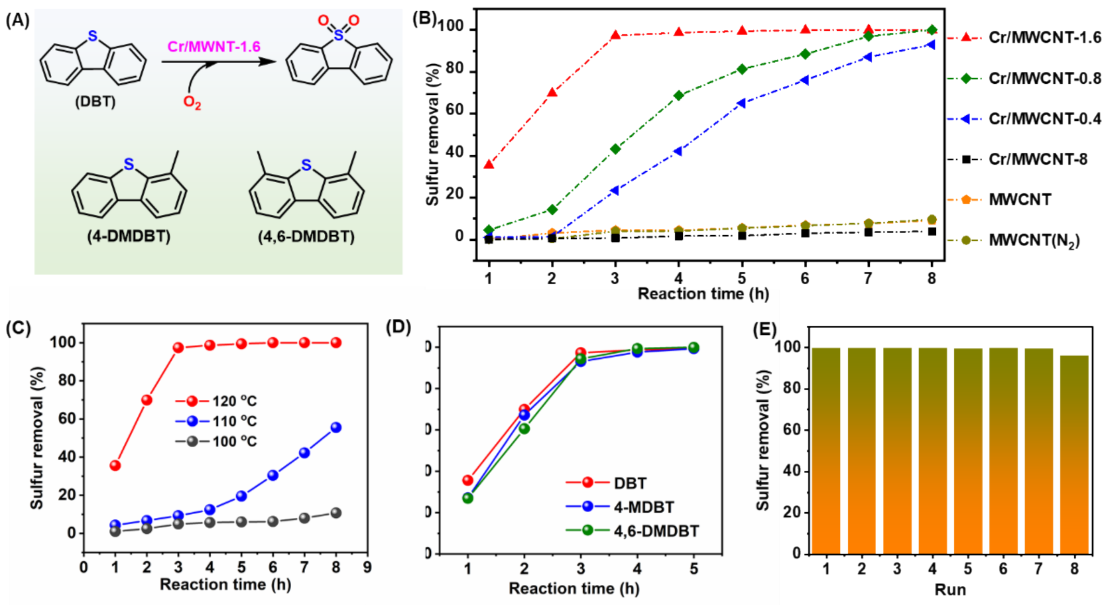

Figure 3. (A) Representative reaction scheme for ASCs-to-sulfone transformation and the structure of different substrates. (B) The catalytic ODS performance with different catalysts, (C) at various reaction temperatures, (D) for different sulfur compounds. (E) The cycle performance of the SACs. The optimal reaction conditions: m (catalyst) = 0.01 g, T = 120 °C, V(model oil) = 20 mL, *v*(air) = 100 mL/min.

As is known, O_2 in air is the most convenient and environmentally benign oxidant in aerobic oxidation, while it is hard to be activated under mild conditions. Especially, the reaction temperature is one of the most important factors on the activity of the catalyst. As shown in Figure 3C, the Cr/MWCNT-1.6 almost had no catalytic activity at 100 °C. As the reaction temperature increased to 110 °C and 120 °C, the desulfurization efficiency can reach 55.3% and 100%, respectively. Compared with the reaction temperature (≥ 380 °C) of hydrodesulfurization (HDS) in industry, this oxidative desulfurization technology with SACs exhibits promising prospect in regard to energy saving and safety problem. Besides DBT, the other two refractory sulfur compounds (4-MDBT and 4,6-DMDBT) were more difficult to be removed from fuel oil.[41-42] Here, the sulfur removal of 4-MDBT and 4,6-DMDBT could reach as high as 99.3% and 99.9%, indicating that the Cr/MWCNT-1.6 was a competitive ODS catalyst with high efficiency and broad substrate scope (Figure 3D). Separation and recycling of catalysts are the essential parameters for the feasible industrial application of an ODS technology. After each cycle, Cr/MWCNT-1.6 was separated via centrifugation, dried, and used for the subsequent cycle. As it can be seen in Figure 3E, the Cr/MWCNT-1.6 possessed excellent recycling performance and the desulfurization efficiency only slightly decreased to 96.0% in the eighth run, demonstrating the strong binding interaction of the Cr SAs and the C–O bond within the MWCNT architectures. Characterization of the reused Cr/MWCNT-1.6 catalyst revealed that the crystalline structure and surface oxygen-containing moieties were well maintained, and the active sites still existed as highly dispersed Cr SAs (Figure S21A-C). In addition, the Cr content in the reused Cr/MWCNT-1.6 catalyst was similar to the fresh one, and no Cr leaching was detected in the oil phase by ICP-AES, indicating that Cr was strongly coordinated with the MWCNT support during the reaction procedure (Table S5).

To explore the oxidation reaction mechanism, free radical quenching experiments were carried out with tert-butyl alcohol (TBA) and p-benzoquinone (BQ) as hydroxyl radical and superoxide radical scavengers. As shown in Figure 4A, the reaction rate was reduced upon addition of TBA but desulfurization efficiency still reached up to 90.1% after prolonging the reaction time to 8 h. In contrast, the catalytic activity was significantly inhibited after BQ was added into the reaction system and the desulfurization efficiency reached merely 10.8% after reacting for 8 h. Therefore, it was deduced that $O_2^{\bullet-}$ could be the main oxygen active species. To verify this hypothesis, the electron spin resonance (ESR) spectroscopy in the presence the spin trap reagent 5,5-dimethyl-1-pyrrolin-N-oxide (DMPO) were performed (Figure 4B). It can be seen that no free radical signal was generated in absence of the SACs. However, a six-fold peak signal could be detected during reaction, which belongs to the DMPO-$O_2^{\bullet-}$ signal. This is consistent with the results of free radical quenching experiment. Therefore, it can be concluded from the above analysis that $O_2^{\bullet-}$ was the main oxygen active species in the oxidation process.

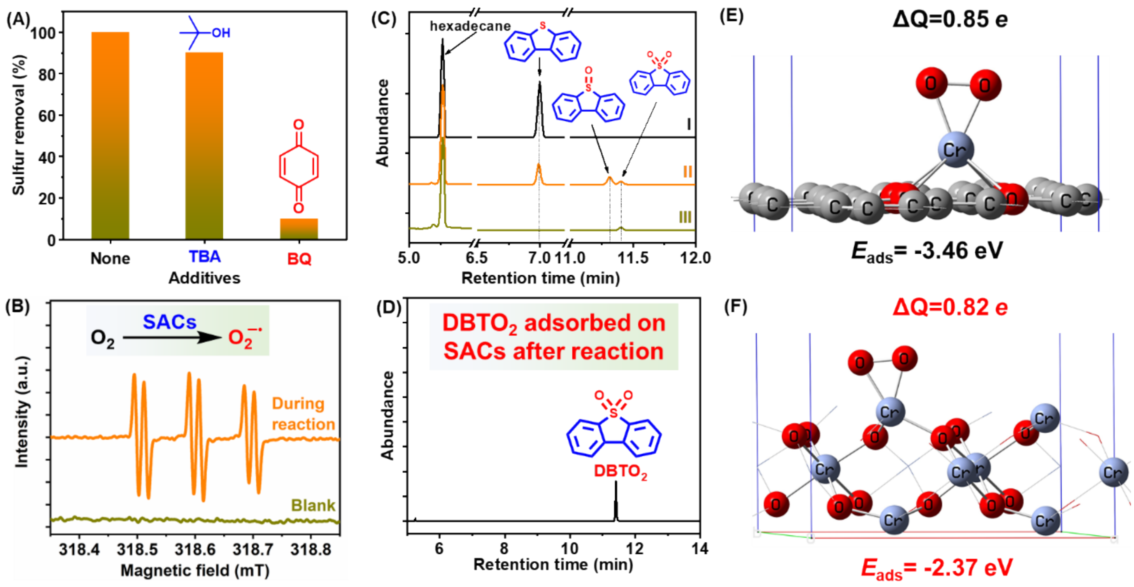

Figure 4. (A) Radical quenching experiments. (B) ESR analysis of active oxygen radicals. (C) GC analysis of oil phase (I) before reaction, (II) after reaction for 1 h, (III) after reaction for 8 h. (D) GC analysis of the catalyst phase (extracted by CCl_4). Optimized structures for the adsorption of O_2 on (E) O-coordinated Cr/MWCNT-1.6

model and (F) Cr_2O_3 (001) surfaces.

In order to reach a deep understanding the role of SA site, periodic density functional theory (DFT) was employed to estimate their adsorptive strength of O_2 and activity to promote the formation of superoxide radical. A graphene model containing Cr SA was constructed to simulate the single atom site of MWCNT surface, which can be served as the active sites for the adsorption of O_2 molecule via the formation of the six-coordinated active intermediates. For comparison, the O_2 adsorption behavior on the bulk Cr_2O_3 surface was also calculated. The optimized configurations for O_2 adsorption on both surfaces are shown in Figure 4E and 4F. It can be seen from Figure 4E that the Cr atom is out of the graphene plane after optimization, and such geometry was preferred for the adsorption of the O_2 molecule on the Cr atom.[43] This result was different from other SA systems consisting of metal sites coordinated by N atoms.[43-44] In addition, the O_2 adsorptive energy (E_{ads}) for this model was −3.46 eV, which was much stronger than the value (E_{ads} = −2.37 eV) obtained on the Cr_2O_3 surface (Figure 4F). The catalytic activity to form superoxide radical was also analyzed by Bader charge method. The results showed that the O_2 molecule could obtain more electrons (ΔQ = 0.85 e) from the Cr SA sites than on the bulk Cr_2O_3 surface, indicating that the O_2 molecule being adsorbed on the Cr SA site was more easily to be activated to form the superoxide radical, leading to enhanced catalytic efficiency in the ODS reaction. Therefore, the theoretical calculations demonstrated that the Cr SA site possessed enhanced O_2 adsorption strength and could facilitate the formation of superoxide radical.

Subsequently, the gas chromatography (GC) analysis of the model oil during the reaction procedure was conducted in three time periods (Figure 4C). Assisted by the mass spectra (MS) (Figure S22), the peak with a retention time of 7 min was belonged to DBT. After reacting for 1 h in the presence of Cr/MWCNT-1.6, the intensity of the DBT peak decreased, while two new peaks with retention time of 11.3 and 11.4 min, respectively, showed up, being assigned to the corresponding oxidative product, sulfoxide (DBTO) and sulfone ($DBTO_2$), respectively. After reacting for 8 h, the peak of DBT completely disappeared, and only a small amount of $DBTO_2$ was observed. These results indicated that DBT was first oxidized to DBTO, which was further oxidized to $DBTO_2$. As only trace amount of $DBTO_2$ was detected in the solution phase, the as-collected solid catalyst was thoroughly washed by carbon tetrachloride (CCl_4) to extract the adsorbed species. GC-MS analysis of the filtrate reviewed that large amounts of the oxidative product $DBTO_2$ was adsorbed by the MWCNT supports (Figure 4D), supplying extra driven-force enabling the substrate-to-product transformation. From the above analysis, it can be concluded that DBT could be adsorbed by the SACs and efficiently oxidized to $DBTO_2$, then producing a low-sulfur fuel. The plausible reaction mechanism was shown in Figure 5.

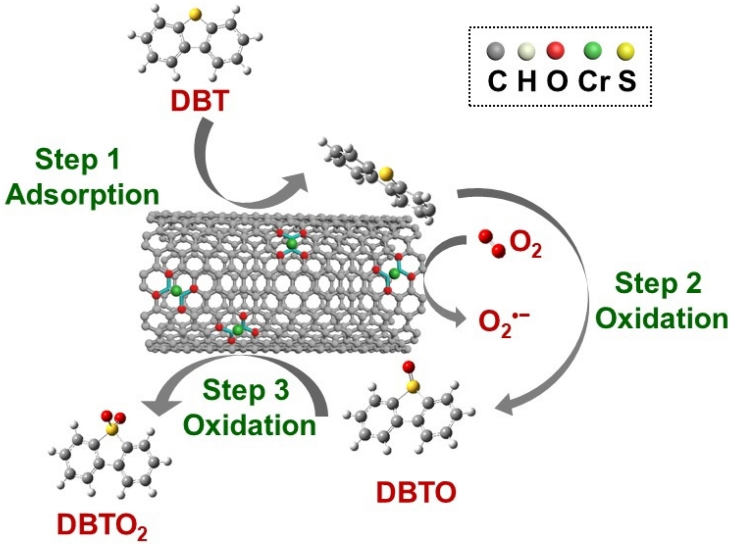

Figure 5. The plausible reaction mechanism of oxidative sulfur removal of DBT catalyzed by Cr/MWCNT catalysts.

In summary, as further efforts to relieve the environmental issues caused by the combustion of sulfur impurities-containing fuels, which played important roles in the fields of catalysis and environmental science, an effective system was developed towards the aerobic ODS transformation. The study revealed that the SACs with earth-abundant metal cores were fabricated via a facile procedure and performed well in the ASC-to-sulfone transformation. Both the metal sites and the unique feature of the support worked synergistically to supply sulfur-free fuels without further purification. This work supplies a promising catalytic system to activate the oxygen molecular and promote the oxidative reaction under mild conditions.

ASSOCIATED CONTENT

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NOTES

The authors declare no competing financial interest.

Supporting Information.

Experimental Section and additional figures and tables.

Table S1. Cr content in the Cr/MWCNT-x catalysts being detected by ICP-AES.

Figure S1. Nitrogen adsorption isotherms of MWCNT collected at 77 K.

Figure S2. The pore size distribution curve of MWCNT.

Figure S3. XRD pattern of MWCNT.

Figure S4. Raman spectra of MWCNT and Cr/MWCNT-1.6.

Figure S5. TEM image of MWCNT.

Figure S6. Nitrogen adsorption/desorption isotherms (77 K) and pore size distribution curve of Cr/MWCNT-8.

Figure S7. O1s XPS spectra of MWCNT and Cr/MWCNT-1.6.

Figure S8. XPS survey spectra of Cr/MWCNT-1.6.

Figure S21. The XRD pattern, O1s XPS spectrum, and HAADF-STEM image of the reused Cr/MWCNT-1.6.

Table S5. Cr content in the fresh and reused catalyst, and the reaction solutions being detected by ICP-AES.

Figure S22. MS analysis of (a) DBT, (b) DBTO and (c) $DBTO_2$.

ACKNOWLEDGEMENTS

This work was financially supported by the National Natural Science Foundation of China (Nos. 21978119, 22078135, and 22178154). SD and ZY were supported by the Division of Chemical Sciences, Geosciences, and Biosciences, Office of Basic Energy Sciences, US Department of Energy.

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